1).

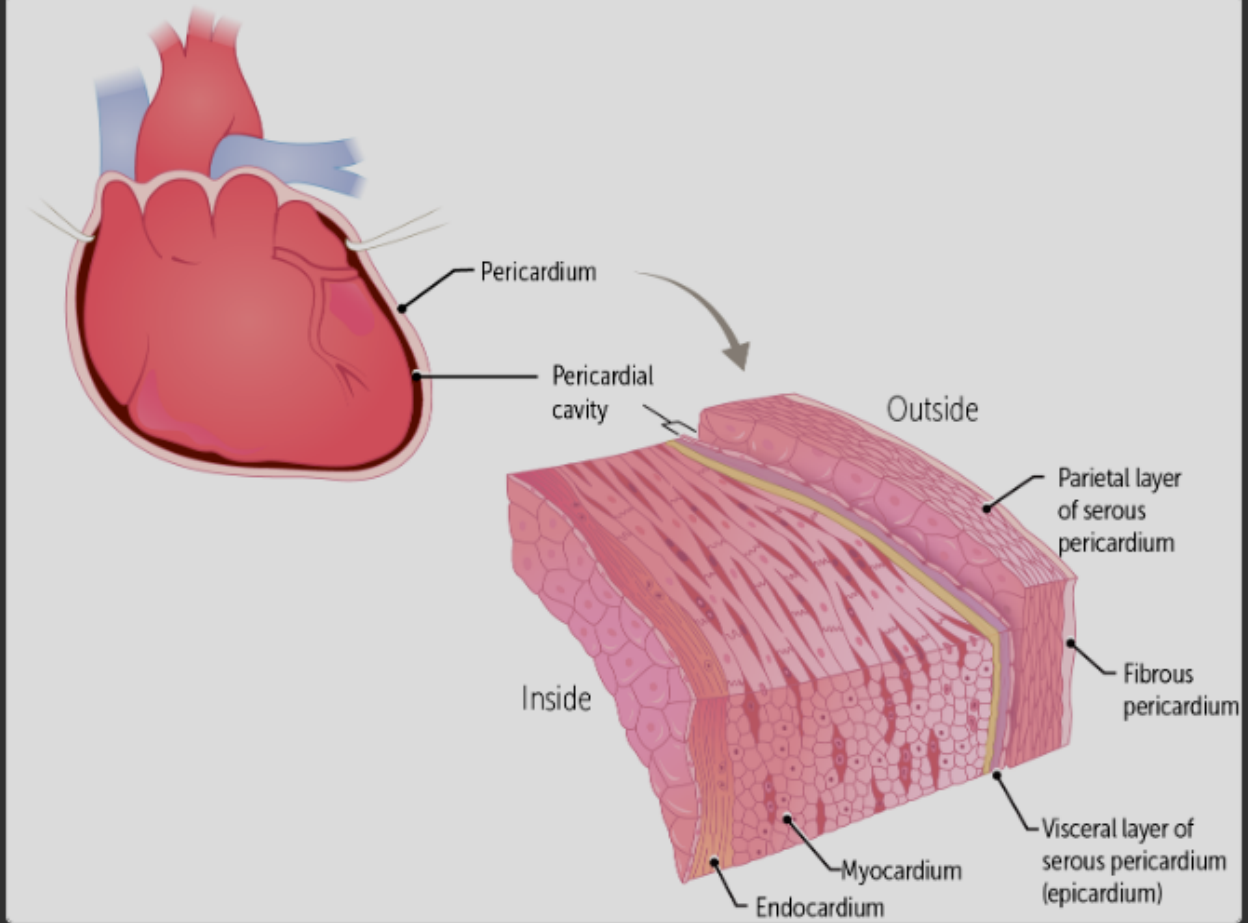

Pericardium
Pericardial cavity
Outside
Inside
Parietal layer of serous pericardium
Fibrous pericardium
Visceral layer of serous pericardium (epicardium)
Myocardium
Endocardium

Figure 1. Structure of the Normal Pericardium.

The pericardium is composed of three layers. From outer to inner, the layers are:

- Fibrous pericardium
- Parietal layer of serous pericardium
- Visceral layer of s

In Figure 2, we can see a histologic view of normal visceral pericardium, which overlays a layer of adipose tissue and myocardium. The visceral layer of the serous pericardium is also known as the **epicardium**.

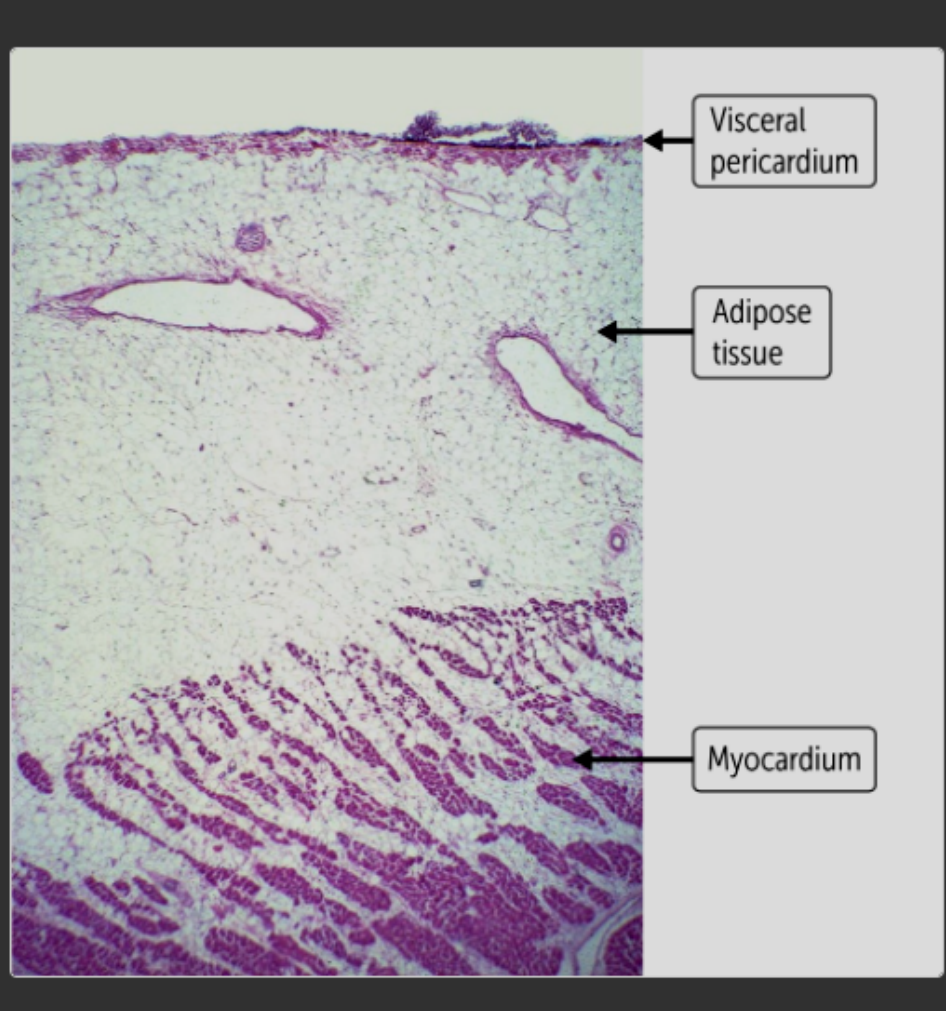


Figure 2. Histology of the Normal Pericardium.

The pericardial cavity is located between parietal and visceral layers of the serous pericardium.

The pericardial cavity normally contains a small amount of pericardial fluid that reduces friction ntraction and relaxation of the heart. When the pericardial tissues become inflamed, the amount of fluid can increase; this is known as a **pericardial effusion**.

What is the definition of acute pericarditis?

How Do Patients With Acute Pericarditis Present? (LO #1)

A patient with acute pericarditis classically presents with acute chest pain that is **pleuritic** in nature (i.e., sharp pain that is worsened by inspiration or coughing).

A classic presentation of the chest pain in acute pericarditis is that the pain improves when the patient is sitting up and leaning forward. Conversely, the chest pain worsens when lying flat. Why does this occur? The pain improves because sitting up and leaning forward reduces the pressure on the parietal layer of the serous pe nsing pain.

Depending on the cause (e.g., infection), there may be associated fever.

What is the distinctive feature of the chest pain in acute pericarditis?

What Is the Pathophysiology of Acute Pericarditis? (LO #2)

Acute inflammation is common to all forms of acute pericarditis, but the individual underlying causes of the inflammation vary and often are challenging to diagnose. Most cases have no clearly identified cause; many of these are presumed to be of viral or autoimmune etiology.

Pathology

Depending on the morphology, acute pericarditis can be classified into four basic pathologic type wever, we rarely categorize patients this way clinically, since biopsy is almost never performed in the management of acute pericarditis.

Fibrinous Pericarditis. Acute fibrinous pericarditis is the most common type of acute pericarditis, and is characterized by the deposition of fibrin on the pericardial surfaces (Figure3).

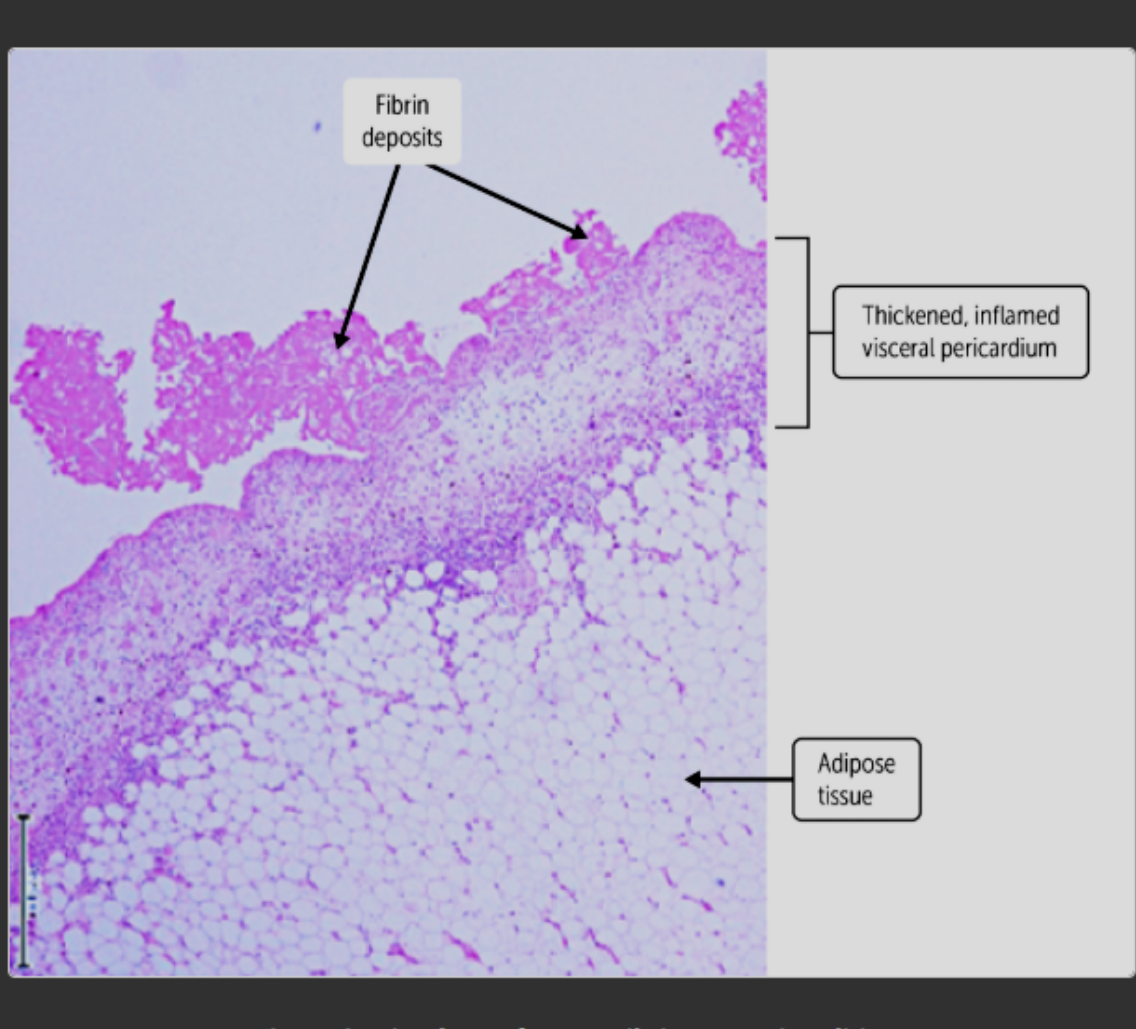


Figure 3. Histology of Acute Fibrinous Pericarditis.

Fibrinous pericarditis may be caused by:

- Myocardial infarc or late

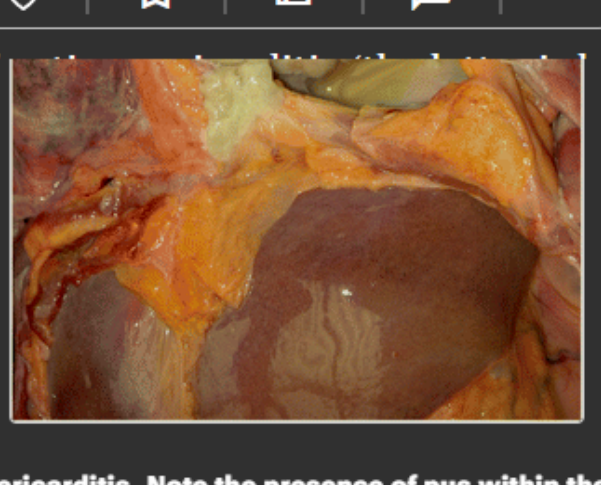


Figure 4. Purulent Pericarditis. Note the presence of pus within the pericardial cavity.

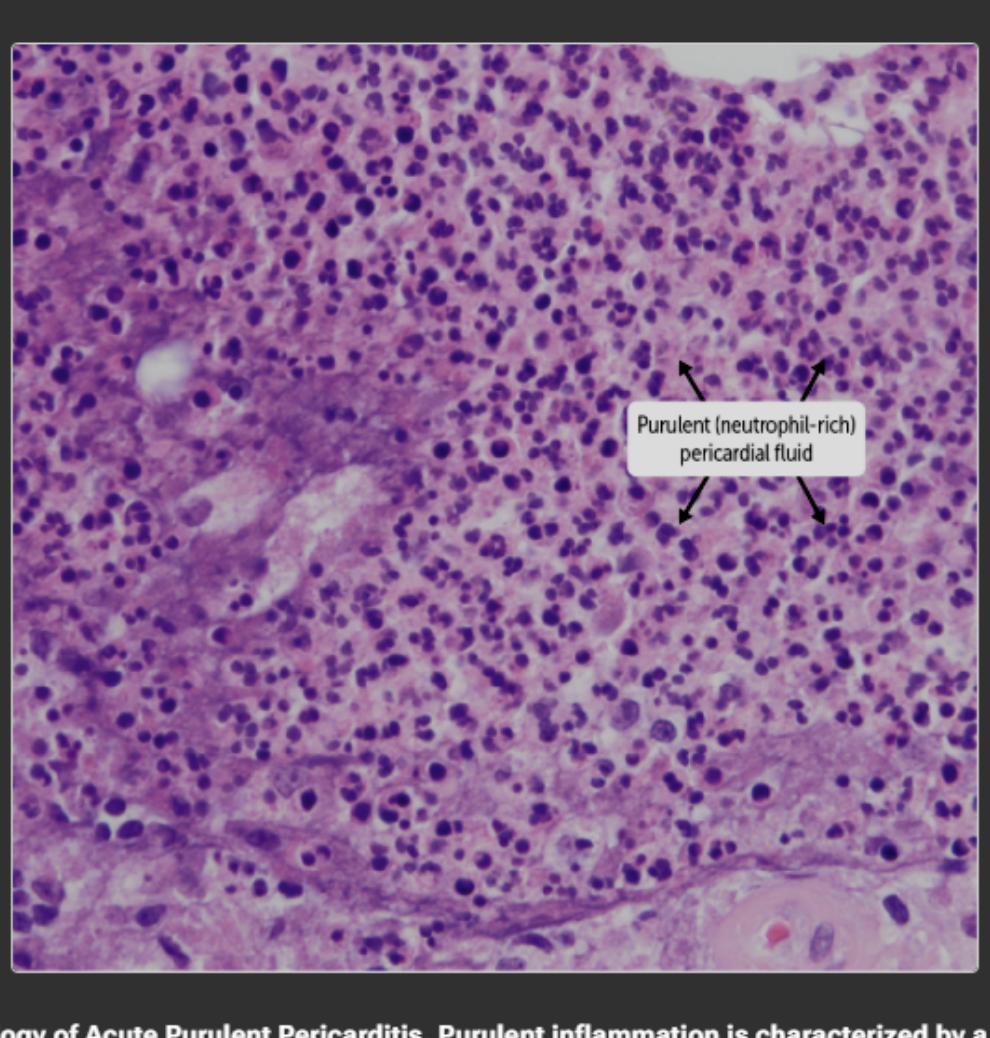


Figure 5. Histology of Acute Purulent Pericarditis. Purulent inflammation is characterized by a neutrophil-rich

How Do We Diagnose Acute Pericarditis? (LO #3)

The four principal diagnostic features of acute pericarditis are chest pain (sharp or pleuritic in nature), a pericardial friction rub, typical ECG findings, and (occasionally) a pleural effusion seen on echocardiogram. For most cases, further workup is usually not indicated unless the patient has suspected bacterial (purulent) pericarditis or cardiac tamponade.

Physical Examination

During the physical exam, a classic finding in acute pericarditis is a pericardial friction rub, though this may not always be present. It is a superficial “scratchy” sound produced when inflamed pericardial layers slide past each other with audible friction that can be detected via auscultation. A friction rub is not always present.

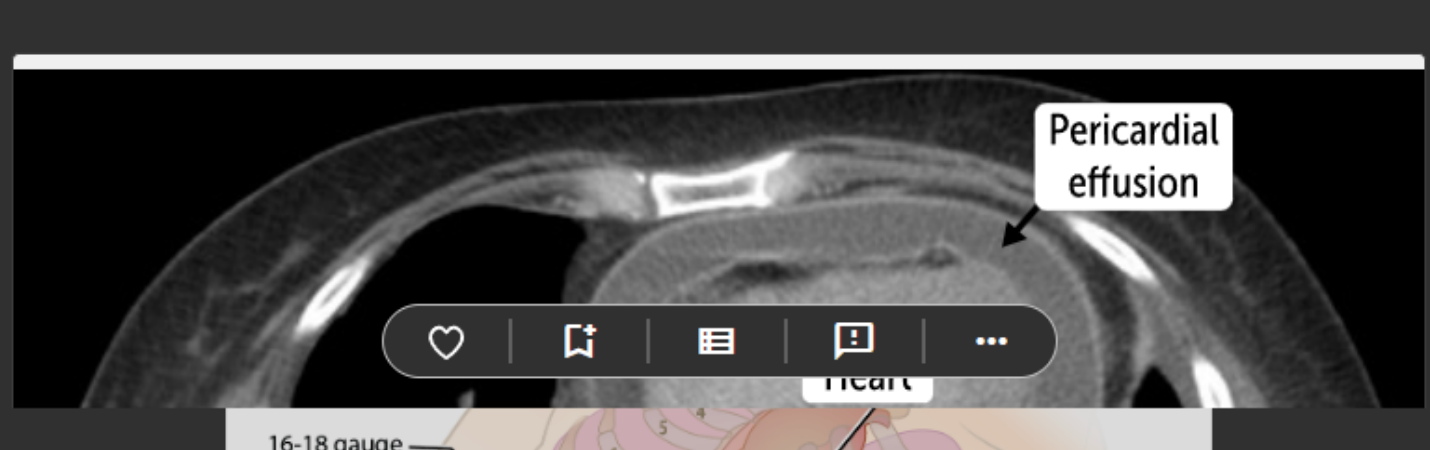
> **CLINICAL CORRELATION**
>
> A pericardial friction rub is a superficial “scratchy” sound produced when inflamed pericardial layers slide past each other with audible friction that can be detected via auscultation. The squeaky leather, sandpaper rubbing,

What are the ECG changes associated with acute pericarditis?

Radiology Studies

We can also utilize radiology studies in evaluating for acute pericarditis:

- Echocardiography can show excess fluid in the pericardial cavity if pericardial effusion is present. However, if no effusion is present, the echocardiogram is typically normal.
- Chest x-ray may be helpful in identifying pulmonary conditions associated with the cause of acute pericarditis (e.g., cancer, sarcoidosis) but is often normal.
- A CT scan can help confirm the diagnosis. It may reveal noncalcified pericardial thickening and, in some cases, a pericardial effusion (Figure 7).
- Cardiac MRI may also be used to detect pericardial inflammation.



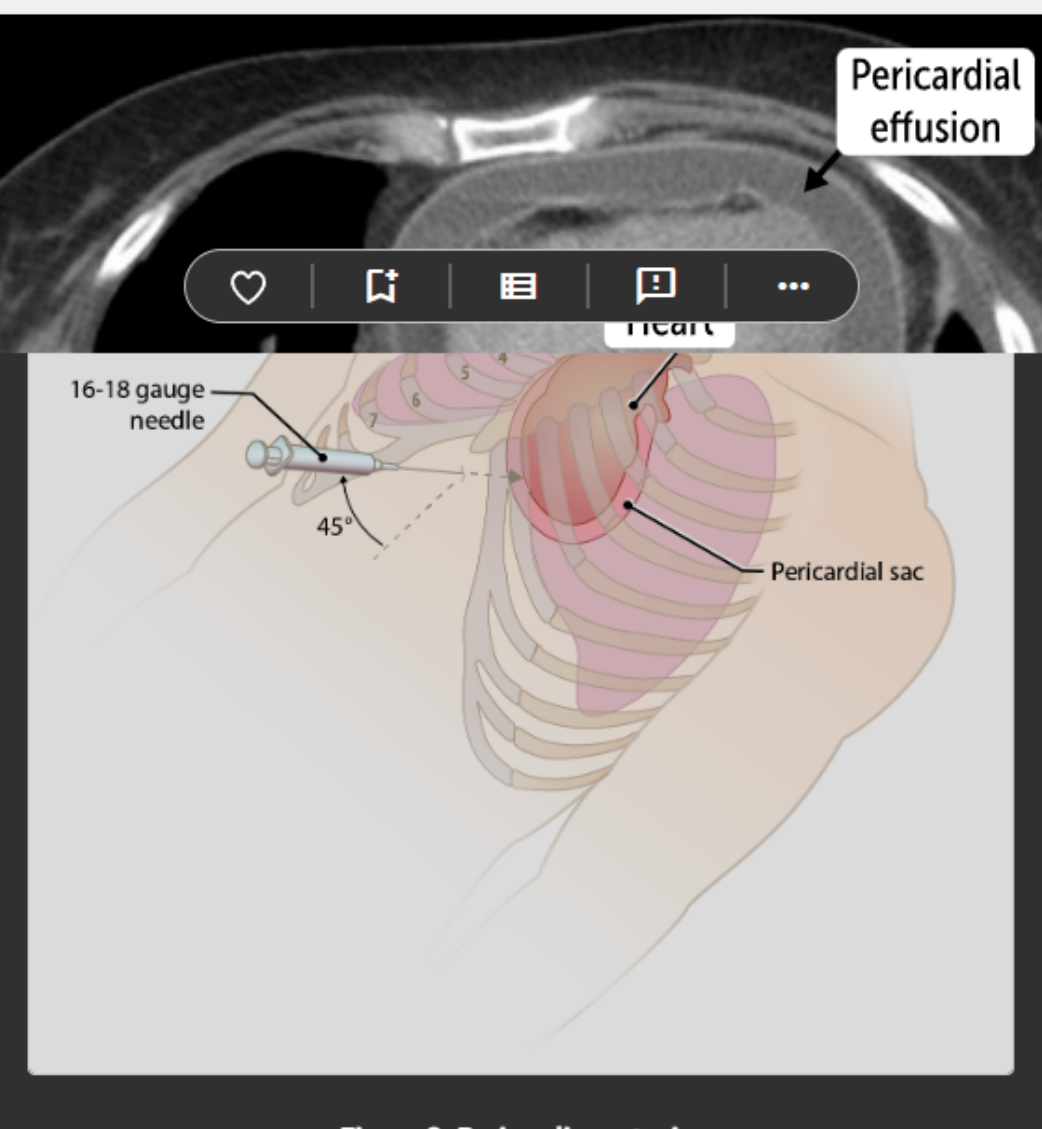


Figure 8. Pericardiocentesis.

How Do We Manage Acute Pericarditis? (LO #4)

The therapeutic goals when managing acute pericarditis are to relieve the chest pain, to resolve the inflammation, and to prevent recurrence. Most patients respond to medications, but a few may require procedural intervention.

Medications

Nonsteroidal anti-inflammatory drugs (NSAIDs) are the primary treatment for acute pericarditis. Colchicine may be considered as an adjunct to NSAID therapy for patients with acute viral or idiopathic pericarditis. Colchicine has also been shown to r d can be used for

> After a review of CT’s ECG and lab results, she is diagnosed with acute pericarditis due to uremia (renal failure). Urgent dialysis is arranged. A few hours later, CT tells you she is feeling much better; her chest pain has improved. You stress the importance of keeping every dialysis appointment to prevent this from happening again. CT understands and agrees to stick to her dialysis schedule.

Summary

- The pericardium is the fibroelastic sac that encloses the heart.
- Acute pericarditis is inflammation of the pericardium that presents suddenly and generally resolves within several weeks.
- Acute pericarditis presents with pleuritic chest pain that is improved by sitting up and leaning forward.
- Common causes of acute pericarditis include infectious, autoimmune, and cardiac etiologies.
- Complications can include pericardial effusion, cardiac tamponade, and constrictive pericarditis.
- A pericardial friction rub may be heard on cardiac auscultation.
- Acute pericarditis can cause diffuse ST elevation and PR depression on ECG.
- Echocardiography can be used to evaluate for the presence of a pericardial effusion.
- Pericardiocentesis can be performed if purulent pericarditis is suspected or in cases of cardiac tamponade.
- Medical managem onsteroidal anti-inflammatory dru r of which can also

A. ○ Coxsackievirus

B. ● Ionizing radiation

C. ○ Aspirin therapy

D. ○ Staphylococcus aureus

Hide Explanation

The correct answer is ionizing radiation (B). Ionizing radiation, particularly following radiation therapy for thoracic malignancies (e.g., breast or lung cancer), can cause fibrinous pericarditis. Coxsackievirus (A) can cause acute serous, not fibrinous, pericarditis. Aspirin (C), an NSAID, is used to treat acute pericarditis; it is not a cause of acute pericarditis. Staphylococcus aureus (D) typically leads to purulent pericarditis.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of acute pericarditis.~~

02 ☑ ~~Explain the pathophysiology of acute pericarditis.~~

03 ☑ ~~Interpret clinical findings to properly diagnose acute pericarditis.~~

04 ☑ ~~Outline the management plan of acute pericarditis.~~

Objective Confidence: 4/4